

MORNING

[Total No. of Questions: 09]

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Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Data Structures

Subject Code: PCIT-101

Paper ID: 16040

Scientific calculator is Not Allowed

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Differentiate between a circular linked list and a linear linked list.
- b) Explain what inserting an element into a queue means.
- c) How is a multi-dimensional array different from a linear array?
- d) What does Big O notation express about an algorithm?
- e) How is a stack represented using a linked list?
- f) Explain the concept of time complexity and why it is a crucial factor in algorithm analysis.

Part – B

[Marks: 04 each]

- Q2.** Explain the key difference between a doubly linked list and a linear linked list. Discuss the operations that can be performed on a doubly linked list.
- Q3.** Explain how the bubble sort algorithm works and discuss its limitations.
- Q4.** Discuss how enqueue and dequeue operations affect the queue structure. How circular queues overcome certain limitations of linear queues?

- Q5. Discuss the advantages and disadvantages of Breadth-First Search (BFS) and Depth-First Search (DFS) in graph traversal. Under what circumstances would you choose one algorithm over the other?
- Q6. Given the infix expression: $(A + B) * C - D / E$, implement a stack-based algorithm to convert it to postfix notation. Provide step-by-step details and the resulting postfix expression.
- Q7. Illustrate the concept of multi-dimensional array and explain its representation. Explain basic operations that can be performed on arrays, such as insertion, deletion, and updating elements with the help of an example.

Part – C**[Marks: 12 each]**

- Q8. Illustrate the concept of bubble sort algorithm in detail with the help of practical example. How does bubble sort works? Explain the advantages and disadvantages of using bubble sort algorithm.

OR

Explain the concepts of tree traversal, specifically in the context of binary trees. Compare and contrast the preorder, inorder, and postorder traversal techniques. Explain traversal techniques with the help of an example.

- Q9. Explain the concept of recursion and its role in problem-solving. Discuss the advantages and challenges of using recursion, using the Tower of Hanoi problem as an illustrative example.

OR

Discuss the fundamental operations that need to be supported for each selected data structure. Explain how these operations are essential for efficient data manipulation in the context of student records.
